MIL-T-47023 (MI)
19 April 1974
SUPERSEDING
MIS-6161
19 September 1968

MILITARY SPECIFICATION

TAPE, HEAT SHRINKABLE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a Polyolefin heat shrink-tape.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-576	Box, Wood, Cleated, Venear, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triple Wall

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

Military

MIL-P-116	Preservation, Methods of
MIL-C-45662	Calibration Systems Requirements

STANDARDS

Federal

FED-STD-228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-104	Limit for Electrical Insulation Color
MIL-STD-810	Environmental Test Methods

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Methods of Test for
ASTM D 150	A-C Loss Characteristics and Dielectric Constant of Solid Electrical Insulating Material, Method of Test for
ASTM D 257	Electrical Resistance of Insulating Materials, Method of Test for
ASTM D 570	Absorption of Plastics, Method of Test for
ASTM D 638	Tensile Properties of Plastics, Methods of Test for
ASTM D 792	Specific Gravity of Plastics by Displacement, Methods of Test for

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacturing of the tape shall be such as to assure performance of the tape in accordance with the requirements of this specification.

3.2 Finished tape.

3.2.1 Rolls. The tape shall be evenly and uniformly wound on cores made of paper fiber or plastic. The core shall have sufficient rigidity to prevent distortion of the roll under normal conditions of transportation and use. When the roll is unwound, the tape shall not tear or split from the face of the tape to the back of the adjacent layer.

3.2.2 Dimensions. The length of the roll shall be 36 yards (108 feet.) The width shall be 0.75 plus or minus 0.05 inches. The thickness shall be 0.020 plus or minus 0.005 inches.

3.2.3 Splices. No splices will be permitted in any roll.

3.2.4 Adhesive. The adhesive shall be pressure sensitive, water insoluble and shall require no moisture, heat or other preparation prior to its application to clean dry surfaces. The coating shall be coated in a smooth manner on one side of the tape.

3.3 Color. When tested as specified in 4.7.2, the color shall be black.

3.4 Dielectric constant.

3.5 Dielectric strength. When tested as specified in 4.7.4, the dielectric strength shall be 1500 volts per mil minimum.

3.6 Volume resistivity. When tested as specified in 4.7.6, the specific gravity shall be 1.3 maximum.

3.7 Specific gravity. When tested as specified in 4.7.6, the specific gravity shall be 1.3 maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

3.8 Tensile strength. When tested as specified in 4.7.7, the tensile strength shall be 1500 pounds per square inch minimum.

3.9 Elongation. When tested as specified in 4.7.8, the elongation shall be 200 percent.

3.10 Flammability. When tested as specified in 4.7.9, for flammability, the tape shall not burn the indicator paper and the flame shall self extinguish itself within 3 inches.

3.11 Water absorption. When tested as specified in 4.7.10, the water absorption shall be 0.1 percent maximum.

3.12 Corrosion. When tested as specified in 4.7.11, the tape shall show no areas of transparency larger than pin holes when examined against a white background. Discoloration is not a corrosive effect and is permissible.

3.13 Fungus. When tested as specified in 4.7.12, the tape shall not support a fungus growth.

3.14 Workmanship. The tape shall contain no foreign particles. The tape edges shall be straight, true and contain no nicks or gouges. The quality of the tape shall meet the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of the inspection equipment shall comply with the requirements of Specification MIL-C-45662.

4.2 Classification of examinations and tests. The examination and testing of coating materials shall be classified as follows:

a. Qualification inspection.

b. Quality conformance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

4.3 Inspection lot. The inspection lot shall consist of all tape of the same type, class and color manufactured by the same process, from the same components, and offered for inspection at one time.

4.4 Standard test conditions. Unless otherwise specified, all tests and inspections shall be made at the standard test conditons as defined below.

a. Temperature, plus 77 plus or minus 9 degrees Fahrenheit (F).

b. Pressure, standard atmosphere.

c. Relative humidity, 60 percent maximum.

4.5 Qualification inspection. Qualification will be performed at a laboratory designated by the government.

4.5.1 Test routine. Qualification inspection shall consist of the examinations and tests specified in Table I.

4.5.2 Sample. A sample shall consist of one roll of tape randomly selected from the lot present for qualification. A minimum of five samples shall be taken. Sample specimens requiring shrinkage shall be made using a suitable heat gun (see 6.3) at a temperature of 750 degrees Fahrenheit (F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

Table I. Qualification Inspection

Examination or Test	Requirement Paragraph	Methods Paragraph
Visual-Mechanical	3.2	4.7.1
Color	3.3	4.7.2
Dielectric Constant	3.4	4.7.3
Dielectric Strength	3.5	4.7.4
Volume Resistivity	3.6	4.7.5
Specific Gravity	3.7	4.7.6
Tensile Strength	3.8	4.7.7
Elongation	3.9	4.7.8
Flammability	3.10	4.7.9
Water Absorption	3.11	4.7.10
Corrosion	3.12	4.7.11
Fungus	3.13	4.7.12

4.6 Quality conformance inspection. Quality conformance inspection shall be performed by the supplier.

4.6.1 Test routine. Quality conformance inspection shall consist of the examinations and tests specified in Table II and shall be performed on each lot offered for inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

4.6.2 Sample. A sample shall consist of one roll of tape randomly selected from the lot. A minimum of two samples shall be taken from each lot offered for inspection.

4.6.3 Failure. Failure of any sample to meet the requirements of Table II shall be cause for rejection of the lot. The supplier shall take corrective action as necessary to preclude recurrence of the failure in subsequently manufactured lots. Acceptance of subsequently manufactured lots may be withheld until the corrective action has been taken.

Table II. Quality Conformance Inspection

Examination or Test	Requirement Paragraph	Methods Paragraph
Visual Mechanical	3.2	4.7.1
Color	3.3	4.7.2
Dielectric Strength	3.5	4.7.4
Tensile Strength	3.8	4.7.7
Elongation	3.9	4.7.8
Flammability	3.11	4.7.9

4.6.3 Test data. The supplier shall maintain all inspection data by lot on all lots offered for acceptance. This data shall be supplied to the procuring agency when requested.

4.6.4 Disposition of samples. Units subjected to the tests shown in Table II shall not be shipped on contract or order.

4.7 Methods of examination and tests.

4.7.1 Visual-mechanical. One yard shall be taken from each roll and examined for holes, tears, cuts or creases. The edges shall not be cut, nicked or gouged. The adhesive coating shall be continuous and free from foreign particles. The rolls shall be evenly and uniformly wound. The tape shall unroll evenly and uniformly and contain no splices. The dimensions shall be as specified in 3.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

4.7.2 Color. The color shall be examined as specified in MIL-STD-104.

4.7.3 Dielectric constant. The dielectric constant shall be tested as specified in ASTM D 150. Wrap tape around a mandrel of copper wire suitable for handling. Shrink tape with heat gun. Apply aluminum foil electrodes over tape.

4.7.4 Dielectric strength. The dielectric strength shall be tested as specified in ASTM D 149. Type of test required is step by step.

4.7.5 Volume resistivity. The volume restivity shall be tested as specified in ASTM D 257. Wrap tape around a suitable mandrel and heat shrink with a heat gun.

4.7.6 Specific gravity. The specific gravity shall be tested as specified in ASTM D 792.

4.7.7 Tensile strength. The tensile strength shall be tested as specified in ASTM D 638. Sample shall be strips cut four inches long by 1/4 inch wide. Testing speed of 20 inches per minute.

4.7.8 Elongation. The elongation shall be tested as specified in ASTM D 638. Sample shall be strips cut four inches long by 1/4 inch wide. Testing speed of two inches per minute.

4.7.9 Flammability

4.7.9.1 Test No. 1. The flammability shall be tested as specified in federal specification FED-STD-228, Method 5221. Tape shall be shrunk over a 1/2 inch mandrel.

4.7.9.2 Test No. 2. Five specimens of tape shall be shrunk over a 1/2 inch steel mandrel for greater than five inches and marked one inch from each end. Each specimen is mounted at a 45 degree angle in a draft free chamber by a suitable clamp. A one inch blue flame is then applied to the lower end of each specimen. When the specimen has burned to the lower mark, the flame is removed. Failure of more than one specimen to extinguish within three inches shall be considered cause for rejection.

4.7.10 Water absorption. The test for water absorption shall be as specified in ASTM D 570 Procedure A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

4.7.11 Corrosion. Specimens shall be one inch by 1/4 inch. The specimen shall be placed in cylindrical glass tubes with a diameter selected to pass the smaller dimension freely. A copper glass mirror is suspended in the tube above the specimen. The mirrors are vacuum deposited copper with a thickness equal to 10 plus or minus 5 percent transmission of normal incident light of 5,000 angstroms. The tubes are sealed tightly with corks wrapped in aluminum foil and the lower end placed in an oven or oil bath at a temperature of 121 degrees plus or minus 2 degrees Celsius (C) for a minimum of 16 hours. The copper mirrors should be more than four inches from the heat source. The mirrors shall be cooled and examined for corrosion.

4.7.12 Fungus. The test for fungus shall be in accordance with MIL-STD-810, Method 508.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (sse 6.2).

5.1.1 Level A. Rolls of tape shall be individually wrapped in Kraft paper conforming to UU-P-268, and sealed in waterproof bags.

5.1.2 Level C. Rolls of tape shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Tape packaged as specified, shall be packed in containers conforming to class weather resistant, PP-B-636, Class 3, PPP-B-585, and Class 2 of PPP-B-640, PPP-B-576, PPP-B-591, and PPP-B-621. The gross weight of each pack shall be limited to 200 pounds with the exception of PPP-B-636 which shall not exceed weight limitations of the applicable container specification.

5.2.2 Level C. Tape packaged as specified, shall be packed in containers acceptable to the common carrier and which will insure delivery at the destination in a satisfactory condition at the lowest applicable rate. Containers or method of packing shall comply with Uniform Freight Classification Rules or other carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking. All containers shall be marked in accordance with MIL-STD-129. In addition, each container shall be marked with:

a. Width of tape.

b. Length of tape.

c. This specification number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The tape is intended as a wrapping around cable, wire bundles and the like. When properly applied, it forms a snug mechanical fit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing (see 5.1 and 5.2).

c. Test data requirements (see 4.6.3).

6.3 Wrapping procedure. The recommended wrapping should be a 1/3 wrap over the previous layer. A 1/3 overlay wrap provides a tight initial fit and when properly heated will cause the material to flow evenly into voids.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-6161, dated 19 September 1968.

Custodian:
ARMY-MI

Preparing Activity
ARMY-MI
Project No. 9320-A180

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974 -713-153/5109

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:35Z
Check the source to verify that this is the current version before use.